

Rimsha Mahmood

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PROFESSIONAL SUMMARY

Data Science undergraduate at NUST (SEECS) with rigorous hands-on experience in **Generative AI**, **Deep Learning**, and **Business Intelligence**. Proven track record of architecting **Graph-based Multi-Agent Systems** and building end-to-end ML pipelines with interactive dashboards in **Power BI** and **Tableau**. Skilled in deploying AI models via APIs and leveraging attention mechanisms for strategic decision-making.

EDUCATION

National University of Sciences & Technology (NUST-SEECS)

Islamabad, Pakistan

Bachelor of Science in Data Science

2023 – 2027

- **Relevant Coursework:** Deep Learning, Natural Language Processing, Artificial Intelligence, Machine Learning, Data Visualization, Data Mining, Database Management (DBMS), Algorithms.

TECHNICAL SKILLS

Generative AI & Agents: LangGraph, Multi-Agent Systems, RAG Pipelines, DeepSeek-R1, Llama-3, Prompt Engineering, Vector Databases (Chroma), RAGAS Metrics

Machine Learning & AI: PyTorch, YOLOv8, Scikit-learn, Bi-LSTM, Attention Mechanisms, OpenCV, SMOTE, Random Forest, XGBoost

Visualization & BI: Power BI (Data Modeling, DAX), Tableau, Matplotlib, Seaborn

Languages: Python (Advanced), SQL (Advanced), Java, DAX, Bash, C++

Data Engineering: Pandas, NumPy, ETL Automation, Data Cleaning/Imputation, Feature Engineering, EDA

Tools & Frameworks: Ollama, Docker, FastAPI, Git, GitHub, SQLite, Google Colab

PROJECTS

Autonomous Academic Research Agent | *LangGraph, DeepSeek-R1, Ollama, arXiv API* Mar 2026 – Present

- Architected a **multi-agent orchestration system** using **LangGraph** to automate the end-to-end synthesis of technical research papers from a single user query.
- Engineered a specialized agent cluster: **Planner** (outline generation), **Researcher** (arXiv filtering), **Writer** (prose synthesis), and **Synthesizer** (final report assembly).
- Integrated **Gamma** for complex reasoning and evidence extraction across 20+ sources, ensuring technical accuracy and consistent in-text citation grounding.
- Built a custom **LLM Factory** to manage local inference via **Ollama**, optimizing model selection between **Llama-3.2** and **Gamma** for specific task requirements.

Automated Equine Pain Recognition | *PyTorch, YOLOv8, FastAPI, Bi-LSTM* Oct 2025 – Dec 2025

- Developed an end-to-end Deep Learning pipeline utilizing **Attention Mechanisms** within a Bi-LSTM architecture, increasing Macro-F1 score by **6.4% (0.801)** over baselines.
- Engineered a high-precision **YOLOv8** system to isolate Regions of Interest (Chin & Eye), achieving **mAP@0.5 of 0.948**.
- Applied **RIFE interpolation** to upsample video data from 7 to 50 FPS and extracted 512-D features using **I3D (ResNet-18)**.
- Optimized inference speed to **125ms per image** via a **FastAPI** interface to enable real-time analysis.

Global Terrorism Intelligence Platform | *Power BI, Tableau, SQL, Python* Oct 2025 – Dec 2025

- Architected a comprehensive analytics solution utilizing the GTD dataset (**138,000+ incidents**) and World Bank data to identify macro-security trends.
- Built a robust **ETL pipeline** in Python to handle missing values, standardize geopolitical nomenclature, and normalize casualty metrics.
- Trained **Random Forest** and **XGBoost** classifiers to predict attack severity and performed feature importance analysis.
- Designed executive-level **Power BI dashboards** using complex **DAX measures** for regional drill-down analysis.

EXPERIENCE

Data Science Intern

Remote

Excelerate

2024

- Conducted large-scale **Exploratory Data Analysis (EDA)** to identify anomalies and improve data quality for downstream modeling.
- Collaborated with the engineering team to validate model performance and implement reproducible ML workflows.

LEADERSHIP & CERTIFICATIONS

Executive HR (NUST EFX & NCF): Managed a team of 50+ members; demonstrated strong stakeholder communication and ownership.

Certifications: SQL Intermediate (HackerRank), Python for Data Science (DataCamp).